Layer 3: Mass Transport in the Incubation Chamber

Diffusion, the Damköhler Criterion, and In-Device Inhibitor Depletion

Eton iGEM 2026, Dry Lab | Layer 3 of 17-Layer Biosensor Pipeline | `diffusion_pde.py`, `damkohler.py`, `fragment_clearance.py`

Once the chamber fills (Layer 2), the device is static: there is no convection, and every enzyme molecule that reaches the electrode surface gets there by diffusion alone. This layer answers the question the whole assay depends on: does the measured signal report the biochemistry we care about (kcat/Km of MMP-9 or HNE on the immobilised probe), or does it report something we do not care about (how fast a 92 kDa protein diffuses across a sub-millimetre gap)? Layers 0–2 hand this layer a bulk chamber concentration $[E]_{chamber}$; Layer 3 turns that into $[E](z{=}0, t)$, the enzyme concentration actually available at the electrode at each moment, and hands that profile to the Layer 9 surface kinetics ODE. Section IX summarises the exchange with all fifteen neighbouring layers.

Three sub-models are chained together: (i) diffusion of enzyme from the bulk to the surface, coupled to surface substrate depletion (Sections I–IV); (ii) competitive inhibitor binding *inside* the device, which depletes the free-enzyme pool independently of the electrode (Section V) — this is the single largest correction this layer makes to a naive diffusion-only picture; and (iii) clearance of the cleaved, MB-labelled fragment away from the electrode after reaction (Section VI). Section VII disposes of a possible complication from the fill phase, and Sections VIII–X give the quality checks, outputs, and limitations.

I. Why Diffusion Cannot Be Neglected

Diffusion coefficients from Stokes–Einstein

For a spherical particle of hydrodynamic radius r_h in a fluid of viscosity η, the Stokes–Einstein relation gives

$$D = \frac{k_B T}{6\pi\eta r_h}. \tag{I.1}$$

For a compact globular protein, r_h scales with molecular weight as $r_h \approx 0.066\,\text{nm} \times (M_W/\text{Da})^{1/3}$ (Tyn & Gusek 1990 [1]). This scaling law reproduces the literature diffusion coefficient of HNE (a compact, single-domain serine protease) to within a few percent, but *underestimates* that of MMP-9: MMP-9 carries a catalytic domain, a fibronectin-like insert, and a C-terminal hemopexin domain connected by a flexible linker, so its true hydrodynamic drag is larger than an equal-mass sphere. We therefore use the literature hydrodynamic radius ($r_h \approx 3.5\,\text{nm}$) for MMP-9 rather than the pure scaling-law estimate, and the scaling law directly for HNE and for the small released MB fragment, where no direct measurement is required to get a sub-10% estimate.

Species	r_h (nm)	$D_{25°C}$ (m^2/s)	Source
MMP-9 monomer (92 kDa)	3.5 (lit.)	5.0×10^{-11}	README Layer 0
MMP-9 dimer (184 kDa)	—	$\approx 4.0 \times 10^{-11}$	$D_{\text{mono}}/2^{1/3}$
HNE (30 kDa)	2.06 (scaling law)	9.2×10^{-11}	scaling law
TIMP-1 (21 kDa)	scaling law	$\approx 1.35 \times 10^{-10}$	scaling law
A1AT (52 kDa)	scaling law	$\approx 1.0 \times 10^{-10}$	scaling law
MB fragment (SGIG-PEG2-Lys(MB), ∼1 kDa)	0.66 (scaling law)	3.2×10^{-10}	scaling law

Temperature correction uses Stokes–Einstein directly, $D(T) = D(T_{ref})\,(T/T_{ref})\,(\eta(T_{ref})/\eta(T))$, with the water viscosity taken from the Vogel–Fulcher–Tammann fit $\eta(T) = 2.414 \times 10^{-5} \times 10^{247.8/(T-140)}\,\text{Pa s}$ (T in Kelvin), which reproduces the tabulated values at 20, 25 and 37 °C to better than 0.2%. Over the device's 20–37 °C operating range this changes D by roughly +30%, and must be tracked alongside k_{cat}'s own (larger) Arrhenius temperature dependence when comparing measurements taken at different temperatures.

Vertical mixing time: the central physical fact of this layer

The relevant question is not whether the enzyme *can* reach the surface, but whether it does so *fast enough* relative to the 30-minute incubation. The characteristic time to equilibrate a concentration step across the channel height h by pure diffusion is

$$t_{mix} \approx \frac{h^2}{2D}. \tag{I.2}$$

For the design midpoint $h = 375\,\mu\text{m}$ and MMP-9,

$$t_{mix} = \frac{(3.75 \times 10^{-4}\,\text{m})^2}{2 \times 5.0 \times 10^{-11}\,\text{m}^2/\text{s}} \approx 1.4 \times 10^3\,\text{s} \approx 23\ \text{min}. \tag{I.3}$$

This is comparable to, not much smaller than, the 30-minute incubation. *The chamber is not well-mixed during the assay.* For HNE (D roughly 1.8× larger), $t_{mix} \approx 13\,\text{min}$ — still not negligible. This single fact is what makes the rest of this layer necessary: a zero-dimensional model that simply assumes $[E]$ at the electrode equals the bulk concentration the whole time is not a safe simplification here, and the correct treatment requires solving the spatial diffusion problem explicitly, including a depletion layer that grows at the electrode as cleavage proceeds.

II. Governing Equations: The Coupled PDE–ODE System

The diffusion PDE and its boundary conditions

With $z = 0$ at the electrode and $z = h$ at the PDMS lid, the enzyme concentration field obeys

$$\frac{\partial [E]}{\partial t} = D_E \frac{\partial^2 [E]}{\partial z^2}, \qquad z \in (0, h),\ t > 0, \tag{II.1}$$

with initial condition $[E](z, 0) = [E]_{chamber}$ (uniform; justified in Section VII), a no-flux condition at the lid,

$$\left.\frac{\partial [E]}{\partial z}\right|_{z=h} = 0, \tag{II.2}$$

and a Robin (radiation) condition at the electrode that couples the diffusive flux arriving at the surface to the rate at which the immobilised probe is being consumed there,

$$\boxed{D_E \left.\frac{\partial [E]}{\partial z}\right|_{z=0} = J_{surface}(t).} \tag{II.3}$$

Closing the system: the linear surface-kinetics regime

Layer 9 supplies $J_{surface}(t)$ from Michaelis–Menten kinetics on the surface. The general form saturates as the remaining intact probe density $\Gamma(t)$ is used up; but Layer 13's own working assumption, and the one this pipeline is designed around, is the linear (pseudo-first-order) regime, $\Gamma \ll K_{m,surf}$, in which

$$\frac{d\Gamma}{dt} = -\kappa(t)\,[E](0,t), \qquad \kappa(t) \equiv \frac{k_{cat}}{K_{m,surf}}\,\Gamma(t). \tag{II.4}$$

$\kappa(t)$ has units of a velocity (m/s): it is the areal reaction rate constant seen by an enzyme molecule sitting exactly at the surface, and $k_{cat}/K_{m,surf}$ is the familiar specificity constant reported by protease kinetics assays (e.g. Kridel 2001, in $\mathrm{M^{-1}s^{-1}}$; SI units $\mathrm{m^3\ mol^{-1}s^{-1}}$). We take $J_{surface}(t) = -d\Gamma/dt = \kappa(t)\,[E](0,t)$, so the Robin condition becomes a genuinely linear boundary condition at every instant,

$$D_E \left.\frac{\partial [E]}{\partial z}\right|_{z=0} = \kappa(t)\,[E](0,t), \tag{II.5}$$

with $\kappa(t)$ itself evolving on the (slow, $\gg t_{mix}$ in the reaction-limited design target) timescale set by $\Gamma(t)$. This closes the system exactly, without any dimensional ambiguity: $[E]$ is a volumetric concentration ($\mathrm{mol/m^3}$) throughout, Γ is an areal density ($\mathrm{mol/m^2}$) throughout, and κ carries the unit conversion between them. *A caveat on this choice is given in Section X: treating the flux consumed by the enzyme as numerically equal to the substrate cleavage rate is a modelling simplification (enzymes are catalytic and are not literally removed from solution by turnover), and it is a conservative one — it can only overstate, never understate, surface enzyme depletion.*

III. Numerical Solution: Crank–Nicolson Finite Differences

Explicit stability and the case for Crank–Nicolson

A straightforward explicit (FTCS) scheme is conditionally stable, requiring $\Delta t < (\Delta z)^2/(2D_E)$. Discretising with $N_z = 50$ points ($\Delta z = h/50 = 7.5\,\mu$m for $h = 375\,\mu$m) and $D_E = D_{MMP9}$ gives $\Delta t_{max} \approx 0.56\,$s, so a 1800 s incubation needs $\gtrsim 3200$ explicit steps — computationally trivial, but fragile: halving Δz (finer grid, e.g. to resolve a sharper depletion layer near the surface) quarters Δt_{max}. Crank–Nicolson is unconditionally stable and second-order accurate in time, so grid refinement does not force smaller timesteps; `diffusion_pde.py` uses it throughout.

Discretisation and the lagged PDE–ODE coupling

Writing $r = D_E\Delta t/(2\Delta z^2)$, the interior update is the standard tridiagonal Crank–Nicolson stencil. The two boundaries are folded in with a ghost-node elimination: at $z = h$, $E_{N_z+1} = E_{N_z-1}$ enforces the no-flux mirror condition exactly; at $z = 0$, the Robin condition $(E_1 - E_{-1})/(2\Delta z) = \kappa(t)E_0/D_E$ gives the ghost value $E_{-1} = E_1 - 2\Delta z\,\kappa E_0/D_E$, which substitutes into the interior stencil to produce a modified boundary row with the surface velocity κ folded implicitly into the diagonal coefficient. The full nonlinearity of the coupled system is handled with the lagged (semi-implicit) scheme specified for this layer: at each timestep t_n, κ is evaluated from the already-known $\Gamma(t_n)$ and

held fixed while the (now linear) Crank–Nicolson diffusion step is solved for $[E](\cdot, t_{n+1})$; Γ is then advanced one step using the newly computed $[E](0, t_{n+1})$,

$$\Gamma(t_{n+1}) = \Gamma(t_n) - \kappa(t_n)\,[E](0, t_{n+1})\,\Delta t. \tag{III.1}$$

This avoids ever having to solve a genuinely nonlinear system at each step, at the cost of a one-step lag in κ that is negligible provided $\Delta t \ll$ the timescale over which Γ changes appreciably (always true in the reaction-limited design target, and checked automatically: Γ is clipped at zero and the run is flagged if it depletes within a handful of timesteps). The banded tridiagonal system is solved with `scipy.linalg.solve_banded` at every step.

IV. The Damköhler Number: Reaction-Limited vs. Diffusion-Limited Operation

A dimensionless design criterion

The Damköhler number compares the surface reaction rate to the diffusive supply rate. Using the $\kappa(t)$ derived in Section II, the natural, dimensionless construction is the ratio of the surface reaction velocity (evaluated at $t = 0$, $\Gamma = \Gamma_0$) to the diffusive velocity across the channel, D_E/h:

$$\boxed{Da_0 \equiv \frac{\kappa(0)}{D_E/h} = \frac{k_{cat}}{K_{m,surf}}\,\Gamma_0\,\frac{h}{D_E}.} \tag{IV.1}$$

- $Da_0 \ll 1$: diffusion resupplies the surface far faster than it is consumed; $[E](0,t) \approx [E]_0$ throughout, and the signal reports $k_{cat}/K_{m,surf}$ — the desired, biochemically interpretable regime.
- $Da_0 \gg 1$: the surface reaction outruns diffusion; a depletion layer forms and the observed cleavage rate is capped by how fast enzyme molecules can arrive, not by the intrinsic enzyme kinetics.

Target: $Da_0 < 0.1$ for the design to be safely interpretable (contents.md §3.5).

A noteworthy, slightly counter-intuitive consequence of this construction: Da_0 is *independent of the bulk enzyme concentration* $[E]_0$. Both the maximum surface flux ($\kappa(0)[E]_0$) and the diffusive supply flux ($D_E[E]_0/h$) are linear in $[E]_0$ in this regime, so it cancels from the ratio. Da_0 is a property of the probe design (Γ_0, $k_{cat}/K_{m,surf}$, channel geometry) alone, not of the clinical sample being tested.

Worked example and design implication

Using the C15-substrate literature specificity constant ($k_{cat}/K_m = 1.88\times10^5\,\mathrm{M^{-1}s^{-1}}$, fouling_model.py) with the $\times10$ steric correction from Layer 9 as a working estimate pending wet-lab surface kinetics (pES-01–08), $\Gamma_0 = 5\,\mathrm{pmol/cm^2}$, $h = 375\,\mu\mathrm{m}$, $D_E = D_{MMP9}$:

$$Da_0 = (1.88\times10^4\,\mathrm{M^{-1}s^{-1}}) \times (5\times10^{-8}\,\mathrm{mol/m^2}) \times \frac{3.75\times10^{-4}\,\mathrm{m}}{5.0\times10^{-11}\,\mathrm{m^2/s}} \approx 7. \tag{IV.2}$$

This is *not* in the target $Da_0 < 0.1$ regime: at the literature-analogue specificity constant, the $5\,\mathrm{pmol/cm^2}$ design point is diffusion-influenced. `diffusion_pde.py` confirms this numerically: at the same operating point, the well-mixed (Da $\to$ 0) idealisation predicts 7.1% probe cleavage over 30 min, while the full coupled solve gives only 1.2% — an 84% shortfall relative to the diffusion-free ideal (Fig. 1). This is a legitimate, actionable finding, not a flaw in the device: it means the

assay's usable signal (and hence LOD) will be worse than a naive kinetics-only calculation would suggest, *unless* Γ_0 is reduced, the true surface $k_{cat}/K_{m,surf}$ turns out to be markedly lower than the solution-phase literature analogue (plausible, given steric hindrance), or the incubation is extended.

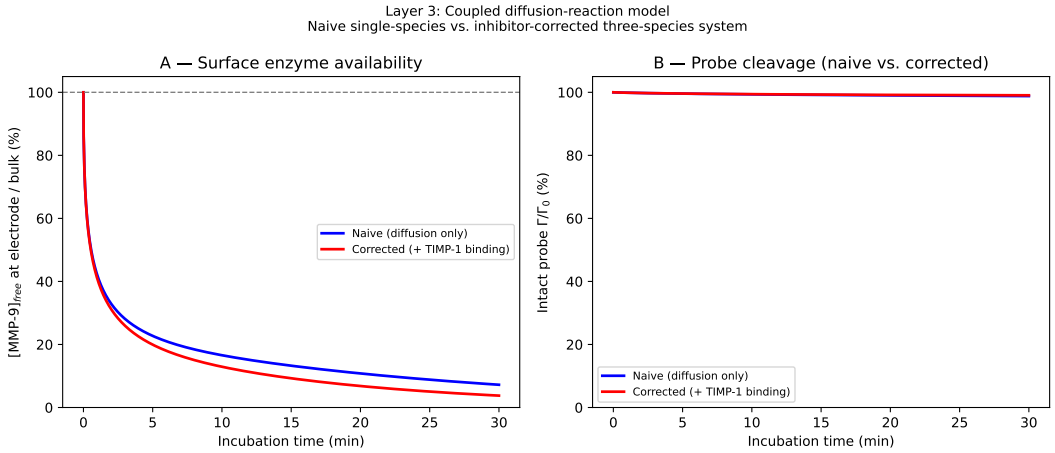


Figure 1: Coupled diffusion-reaction solution at the design operating point ($\Gamma_0 = 5\,\text{pmol/cm}^2$, $[E]_0 = 2.17\,\text{nM}$ MMP-9, C15-analogue $k_{cat}/K_{m,surf}$). **A:** enzyme concentration at the electrode surface, relative to bulk, for the naive (diffusion-only) model and the inhibitor-corrected three-species model. **B:** corresponding intact-probe fraction $\Gamma(t)/\Gamma_0$ for both models (`diffusion_pde.py`).

The apparent Damköhler number: why $[E]_0$ still matters in practice

Da_0 is a fast screening estimate evaluated at $t = 0$; it does not capture how much cumulative signal is actually lost to transport over a finite incubation, because $\Gamma(t)$ and $[E](0,t)$ are coupled *multiplicatively* through $\kappa(t) = (k_{cat}/K_{m,surf})\Gamma(t)$ — this makes the full time-dependent system genuinely bilinear (not linear) in $(\Gamma, [E])$ jointly, even though each individual equation is linear in the sense of having no Michaelis–Menten saturation. Concretely: a higher $[E]_0$ depletes $\Gamma(t)$ faster, which lowers $\kappa(t)$ over the course of the incubation, changing the time-integrated reaction/diffusion balance even though the *initial* balance (Da_0) is unchanged. `damkohler.py` therefore defines a complementary, time-integrated apparent Damköhler number from the full numerical solution,

$$Da_{apparent}(\Gamma_0, [E]_0) = \frac{-\ln(1 - f_{ideal})}{-\ln(1 - f_{actual})} - 1, \tag{IV.3}$$

where $f_{ideal} = 1 - e^{-\kappa(0)[E]_0 T/\Gamma_0 \cdot \Gamma_0}$ is the well-mixed cleavage fraction over the incubation T and f_{actual} is the cleavage fraction from the coupled PDE–ODE solve; $Da_{apparent} \to 0$ recovers reaction-limited behaviour and $Da_{apparent} \gg 0$ flags a diffusion-limited operating point, exactly as for Da_0. Sweeping $\Gamma_0 \in [0.5, 15]\,\text{pmol/cm}^2$ and $[E]_0 \in [0.1, 10]\,\text{nM}$ (Fig. 2) confirms the analytic Da_0 prediction (Fig. 2A: perfectly $[E]_0$-independent contours) alongside the numerical $Da_{apparent}$ map (Fig. 2B), which shows the expected weak but real curvature in $[E]_0$ at fixed Γ_0 from this bilinear coupling. Both maps agree on the qualitative design conclusion: $Da < 0.1$ at this specificity constant requires Γ_0 well below $1\,\text{pmol/cm}^2$, tightening the probe-density window below the steric-accessibility ceiling ($\Gamma_{steric,max} \approx 5\,\text{pmol/cm}^2$, Layer 8) rather than the 1–10 pmol/cm^2 range quoted for enzyme access alone.

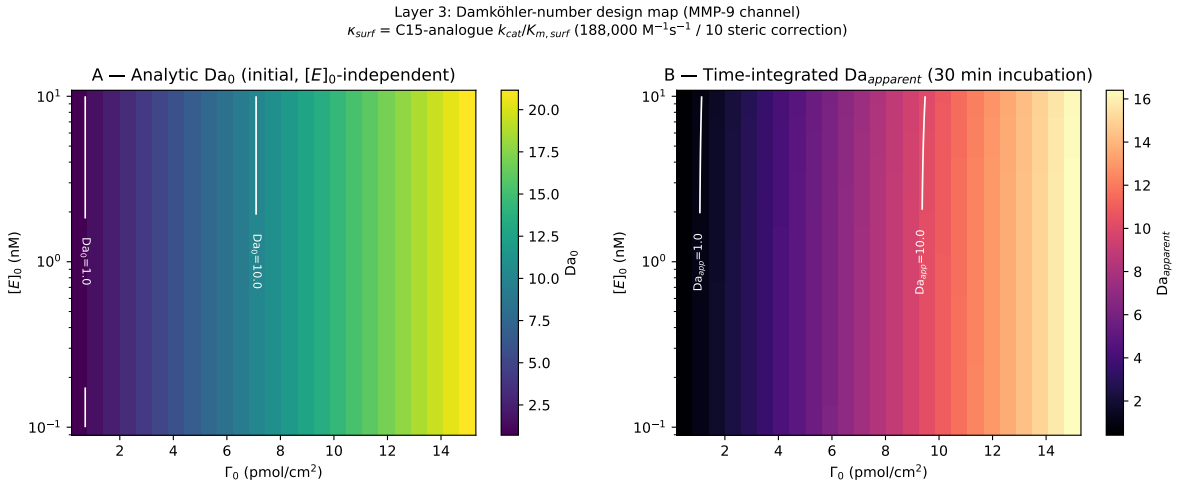


Figure 2: Damköhler design map for the MMP-9 channel (`damkohler.py`). **A:** closed-form Da_0, exactly independent of $[E]_0$ (vertical contours). **B:** time-integrated $Da_{apparent}$ from the full coupled solve over a 30-minute incubation, showing the expected weak dependence on $[E]_0$ from the bilinear Γ-$[E]$ coupling.

V. In-Device Inhibitor Depletion: The Three-Species Correction

Why the pure diffusion PDE is incomplete

Sections I–IV treat $[E]$ as a passively diffusing, non-reacting-in-the-bulk species: its only sink is the electrode. This is wrong for the real assay. TIMP-1 (co-eluted with MMP-9) and A1AT/SLPI (co-eluted with HNE) continue to bind and inactivate free enzyme throughout the incubation, in the bulk solution, independently of anything happening at the electrode. Contents.md identifies this as "the most important addition to this layer identified in lifecycle analysis": omitting it means Layer 9 is fed an enzyme concentration that is systematically too high, biasing every downstream quantity (F_k, κ_k, the calibration curve, the LOD).

The three-species reaction-diffusion system

For the MMP-9/TIMP-1 pair (HNE/A1AT is directly analogous),

$$\frac{\partial [E]_{free}}{\partial t} = D_E \frac{\partial^2 [E]_{free}}{\partial z^2} - k_{on}[E]_{free}[I]_{free} + k_{off}[C], \tag{V.1}$$

$$\frac{\partial [I]_{free}}{\partial t} = D_I \frac{\partial^2 [I]_{free}}{\partial z^2} - k_{on}[E]_{free}[I]_{free} + k_{off}[C], \tag{V.2}$$

$$\frac{\partial [C]}{\partial t} = D_C \frac{\partial^2 [C]}{\partial z^2} + k_{on}[E]_{free}[I]_{free} - k_{off}[C], \tag{V.3}$$

with $[E]_{free}$ alone subject to the Robin sink of Section II at $z = 0$; $[I]_{free}$ and $[C]$ satisfy no-flux conditions at *both* boundaries, since only the free enzyme reacts with the immobilised probe. `diffusion_pde.py` solves this with symmetric operator splitting: an explicit half-step bulk-reaction update (pointwise in z, using the ODE forms of the three equations above with the diffusion terms dropped), a full Crank–Nicolson diffusion step for each species (with the electrode sink applied only to $[E]_{free}$), and a second reaction half-step. Because the reaction terms are evaluated pointwise and explicitly, stability requires only that the timestep be small relative to the fastest pseudo-first-order rate present, which holds comfortably at the grid resolutions used here.

Timescales: why TIMP-1 matters and A1AT matters even more
TIMP-1 is a tight-binding MMP-9 inhibitor ($K_i \sim 0.01\,\text{nM}$, $k_{on} \approx 2 \times 10^5\,\text{M}^{-1}\text{s}^{-1}$). At a representative in-chamber concentration of 2–3 nM TIMP-1, the pseudo-first-order inactivation rate is

$$k_{obs} = k_{on}[TIMP\text{-}1] \approx 2 \times 10^5 \times 2.3 \times 10^{-9}\ \text{s}^{-1} \approx 4.6 \times 10^{-4}\ \text{s}^{-1}, \tag{V.4}$$

a half-life around 25 minutes — comparable to the incubation itself, so a substantial (though not complete) fraction of MMP-9 is captured over 30 min. HNE and A1AT are a starker case: with $k_{on} \approx 2 \times 10^7\,\text{M}^{-1}\text{s}^{-1}$ and A1AT typically present at $\sim 1\,\mu$M, $k_{obs} \approx 20\,\text{s}^{-1}$, a half-life of ~ 35 ms — effectively instantaneous on the assay timescale. *The clinically meaningful HNE signal is therefore whatever free HNE survives the first seconds of A1AT exposure, not total HNE*; the device is, by construction, reporting the free-enzyme fraction that has clinical relevance in wound biology (excess protease that has escaped the body's own inhibitor reserve), which is a feature of the design, not an artefact to be corrected away.

Figure 1 compares the naive (diffusion-only) and inhibitor-corrected surface-availability profiles for the MMP-9/TIMP-1 channel at the design operating point: the free-enzyme concentration at the electrode is suppressed further, and Section 3.9's correction compounds with the diffusion-limitation already identified in Section IV rather than replacing it — both effects act in the same direction, both suppress the achievable signal below the naive kinetics-only estimate.

VI. Released-Fragment Clearance and Cross-Talk

Clearing the Nernst diffusion layer
After cleavage at AAPV↓SGIG, the MB-labelled fragment (SGIG-PEG2-Lys(MB), $D_{frag} = 3.2 \times 10^{-10}\,\text{m}^2/\text{s}$) must leave the thin diffusion layer adjacent to the electrode ($\delta \approx 50$–$100\,\mu$m for a quiescent solution) before the SWV scan, or it contributes a diffusion-controlled peak that distorts the surface-confined peak Layer 12 is trying to extract. The clearance time is

$$t_{clear} = \frac{\delta^2}{D_{frag}} \approx \frac{(50\text{–}100\,\mu\text{m})^2}{3.2 \times 10^{-10}\,\text{m}^2/\text{s}} \approx 8\text{–}31\ \text{s} \tag{VI.1}$$

(`fragment_clearance.py`, Fig. 3A) — fast compared to any realistic delay between incubation and the SWV scan, so this is not a practical concern for a single scan.

Bulk accumulation
Fragment does not disappear; it accumulates in the bulk over the full incubation. For a single 1 mm^2 electrode reaching $F_k = 0.5$ at $\Gamma_0 = 5\,\text{pmol/cm}^2$ in a 37.5 μL chamber,

$$[MB]_{bulk} = \frac{\Gamma_0 A_{elec} F_k}{V_{chamber}} \approx 0.67\ \text{nM}, \tag{VI.2}$$

roughly two orders of magnitude below the ~ 100 nM solution-phase SWV detection floor (Fig. 3B): bulk fragment accumulation is not expected to contribute measurable background, even summed loosely over several electrodes and higher F_k.

Lateral cross-talk between electrodes
Fragment released at one electrode can in principle diffuse laterally and reach a neighbouring electrode running a different probe channel. The characteristic lateral diffusion time over a separation d_{elec} is $t_{cross} = d_{elec}^2/(2D_{frag})$: at the 1 mm design minimum, $t_{cross} \approx 26$ min, comparable to the

incubation; at 2 mm, $\approx$ 104 min, well beyond it (Fig. 3C). A point-source upper bound on the concentration a neighbour would see even at $d_{elec} = 1$ mm and $T = 30$ min (~ 6 nM, using the same $F_k = 0.5$ release as above) remains more than an order of magnitude below the SWV detection floor. Cross-talk is not expected to be a practical concern for the ≥ 1 mm electrode pitch specified in the design, though the margin is comfortable rather than large at exactly 1 mm, and electrode pitch below this should be re-evaluated against this estimate.

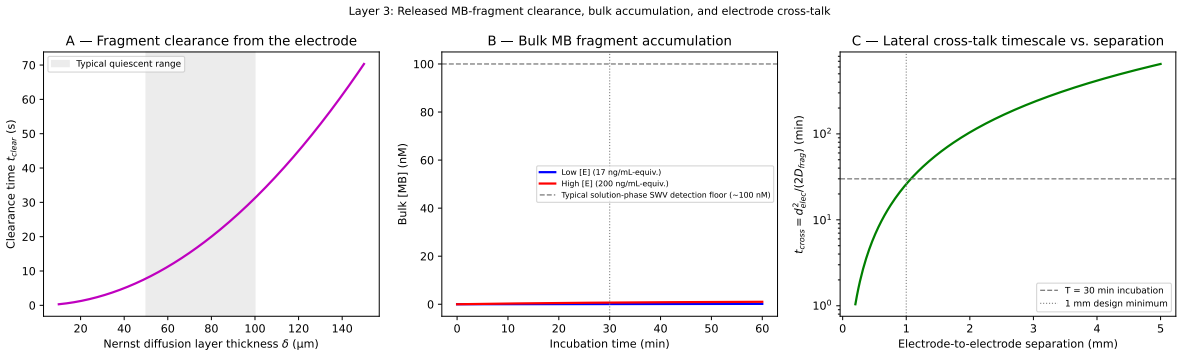


Figure 3: Released MB-fragment behaviour (`fragment_clearance.py`). **A:** Nernst-layer clearance time vs. diffusion layer thickness δ. **B:** Bulk fragment accumulation over a 60-minute incubation for low- and high-clinical enzyme concentrations, compared to the ~ 100 nM solution-phase SWV detection floor. **C:** Lateral cross-talk timescale vs. electrode separation, compared to the 30-min incubation time.

VII. Fill-Phase Convective Uniformity

The uniform initial condition $[E](z, 0) = [E]_{chamber}$ assumed in Section II requires justification: is the enzyme actually uniformly distributed along the channel by the time diffusion becomes the operative transport mechanism? During capillary fill the flow is strongly convective. The Péclet number for MMP-9 at a representative fill velocity $u \approx L/t_{fill}$ ($L = 20$ mm, $t_{fill} \approx 60$ s, so $u \approx 0.33$ mm/s) is

$$Pe = \frac{uL}{D_{MMP9}} \approx \frac{(0.33 \times 10^{-3}\,\mathrm{m/s})(20 \times 10^{-3}\,\mathrm{m})}{5.0 \times 10^{-11}\,\mathrm{m^2/s}} \approx 1.3 \times 10^5 \gg 1, \tag{VII.1}$$

confirming that convection dominates diffusion during fill by five orders of magnitude, so the enzyme is carried uniformly along the channel length essentially instantaneously relative to any diffusive spreading during that same interval. The uniform-IC assumption is safe. (This says nothing about electrode-to-electrode timing offsets along the channel length, which are a separate, smaller effect noted in Layer 2.)

VIII. Quality Diagnostics

Every coupled PDE–ODE run performed by `diffusion_pde.py` is checked against the following automated diagnostics.

Diagnostic	Trigger	Physical interpretation
CFL ratio (info only, CN is unconditionally stable)	$r = D_E\Delta t/(2\Delta z^2) > 5$	Very coarse timestep relative to grid; solution may still be stable but under-resolves the depletion-layer curvature near $z = 0$ — reduce Δt or coarsen Δz to bring r closer to $O(1)$.
Negative concentration clip triggered	$[E]$ or Γ clipped at 0 more than once per run	Timestep too large relative to the local depletion rate; refine N_t.
Γ exhausted early	$\Gamma(t) = 0$ reached before t_{end}	Probe fully consumed; F_k has saturated and the linear/first-order interpretation (Layer 13) no longer applies for the remainder of the run.
Da_0 vs. $Da_{apparent}$ disagreement	$\lvert Da_{apparent} - Da_0\rvert/Da_0 > 50\%$	Strong bilinear feedback between $\Gamma(t)$ depletion and $[E](0,t)$; the closed-form screening estimate is unreliable at this operating point and the full numerical solve should be trusted.
Depletion-layer resolution	surface-adjacent grid spacing $\Delta z >$ estimated depletion-layer thickness $\sqrt{D_E t_{end}}$	Grid too coarse to resolve the profile near $z = 0$; increase N_z.

IX. Layer Outputs

Output	Physical meaning	Typical value	Feeds
$[E](z{=}0,t)$	Free enzyme concentration at the electrode over time	0–100% of bulk	Layer 9
$\Gamma(t)$	Intact surface probe density (already advanced one layer ahead, for validation)	0–Γ_0	Layer 9 cross-check
Da_0	Instantaneous, closed-form Damköhler number	O(0.1–10)	Layer 16 (defensible prediction)
$Da_{apparent}(\Gamma_0, [E]_0)$	Time-integrated Damköhler number, design map	O(0.1–10)	Probe-density design (Layer 7/8)
t_{clear}	MB fragment clearance time from the Nernst layer	8–31 s	Layer 11 (SWV timing)
$[MB]_{bulk}(t)$	Bulk fragment accumulation	sub-nM	Layer 11 (background check)
Confirmation of uniform IC	Pe number for fill-phase convection	$\sim 10^5$	Layer 9

X. Limitations

1. **The enzyme-consumption Robin BC is a conservative approximation.** Section II's boundary condition treats the enzyme flux consumed at the surface as numerically equal to the substrate cleavage rate. Physically, MMP-9 and HNE act catalytically: they are not stoichiometrically removed from solution by cleaving a probe, and should in principle be free to diffuse away and react again. The model as implemented therefore *overstates* surface depletion relative to reality (it behaves as if every cleavage event permanently removes one enzyme molecule from the diffusing pool). This makes the Section IV Damköhler findings a conservative (worst-case) estimate of the diffusion penalty: the true diffusion limitation is expected to be less severe than reported here.
2. **$K_{m,surf}$'s units are only pinned down in the linear regime.** The general (non-linearised) Michaelis–Menten surface ODE quoted in Layer 9's documentation, $d\Gamma/dt = -k_{cat}[E]\Gamma/(K_{m,surf} + \Gamma)$, is ambiguous as written: for the ratio $\Gamma/(K_{m,surf} + \Gamma)$ to be dimensionless, $K_{m,surf}$ must share Γ's areal units (mol/m^2), but Layer 9's own derivation ($K_{m,surf} = K_{m,solution}$/steric factor) treats it as a volumetric solution-phase concentration. This ambiguity is immaterial for every quantitative result in this report, because k_{cat} and $K_{m,surf}$ only ever appear as the lumped ratio $k_{cat}/K_{m,surf}$ (a standard $\mathrm{M^{-1}s^{-1}}$ specificity constant, exactly as reported by wet-lab or literature protease kinetics assays) in the validated linear regime that Layers 9, 13 and 15 are all designed around. It becomes material only if Γ_0 approaches $K_{m,surf}$ (saturating kinetics), which the Section IV design analysis argues against as an operating point in any case. Resolving it fully requires wet-lab surface kinetics data (pES-01–08) reported in explicitly areal units.
3. **The inhibitor reaction-diffusion coupling uses explicit operator splitting.** This is simple and robust at the timesteps used here ($k_{obs}\Delta t \ll 1$ for both TIMP-1 and A1AT/SLPI pathways), but the very fast A1AT/HNE pseudo-first-order rate ($\sim 20\,\mathrm{s^{-1}}$) would require correspondingly small sub-steps if extended to finer time resolution than used here; a fully implicit (or exponential integrator) treatment would be needed for that regime and is not currently implemented.
4. **The lateral cross-talk estimate (Section VI) uses an idealised instantaneous point release.** The true release is continuous and spatially distributed across each electrode's area, not an instantaneous point source; the point-source Green's-function estimate used here is a convenient, order-of-magnitude upper bound rather than an exact calculation, and should be replaced with a full 2-D (or 3-D) finite-difference solve if the electrode pitch is ever pushed below the current 1 mm design minimum.
5. **MMP-9 dimerisation is not currently tracked as a distinct diffusing species in the coupled solver.** `diffusion_pde.py` exposes $D_{MMP9,dimer}$ for use if wet-lab data indicate a non-negligible dimer fraction in wound fluid, but the demonstrations in this report use the monomer diffusivity throughout, consistent with pro-MMP-9 and active MMP-9 being treated as size-equivalent elsewhere in the pipeline.

References

[1] Tyn MT, Gusek TW (1990). Prediction of diffusion coefficients of proteins. *Biotechnol Bioeng* 35:327–338. DOI: 10.1002/bit.260350402 — globular-protein hydrodynamic radius scaling law confirmed.

[2] Squires TM, Messinger RJ, Manalis SR (2008). Making it stick: convection, reaction and diffusion in surface-based biosensors. *Nat Biotechnol* 26:417–426. DOI: 10.1038/nbt1388 — Damköhler-number framework for surface-based biosensor transport limits confirmed.

[3] Beatty K, Bennett B, Otterness I, Travis J (1980). Kinetics of association of serine proteinases with native and oxidized alpha-1-proteinase inhibitor and alpha-1-antichymotrypsin. *J Biol Chem* 255:3931–3934 — HNE/A1AT association rate constant confirmed.

[4] Kridel SJ, Chen E, Kotra LP, Howard EW, Mobashery S, Smith JW (2001). Substrate hydrolysis by matrix metalloproteinase-9. *J Biol Chem* 276:20572–20578. DOI: 10.1074/jbc.M100900200 — MMP-9 specificity constant on the A11/SGKIPRTLTA analogue substrate confirmed.

[5] Bard AJ, Faulkner LR. *Electrochemical Methods: Fundamentals and Applications*, 2nd ed. Wiley, 2001 — finite-difference treatment of diffusion coupled to surface reactions confirmed.